Paper 66 — Analyzing Sharpness along GD Trajectory Progressive Sharpening and Edge of Stability

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The paper in one minute

Question: Why does Hessian sharpness rise toward $2/\eta$ and then oscillate near that value while full-batch GD still makes progress?
Model/Method: Four-phase analysis of GD trajectories, experiments on neural networks, and conditional theorems for a two-layer *linear* network.
Finding: Growth of an output-weight norm is correlated with progressive sharpening; negative feedback from an unstable top residual mode can reduce sharpness after overshooting $2/\eta$.
Why we care: Explains edge-of-stability (EoS) as a moving interaction between curvature and discrete updates, not a constant-curvature theorem.

1. The puzzle: a loss can improve even when the local quadratic picture says "unstable"

Sharpness means the *largest Hessian eigenvalue* $\Lambda(t) = \lambda_{\max}(\nabla^2 L(\theta_t))$. For a fixed positive quadratic direction, GD is locally stable only when $\eta\Lambda < 2$. Yet Figure 1 shows neural-network training continuing after Λ reaches roughly $2/\eta$. Why? The parameters move, so the Hessian and the meaning of "locally unstable" change every step. The paper follows those changes rather than freezing curvature.
Four phases (Section 3.1). I: $\Lambda < 2/\eta$ and progressive sharpening; II: $\Lambda > 2/\eta$ and a top residual component grows; III: curvature drops while that component may still grow; IV: sharpness falls below the threshold, the residual contracts, and the cycle can repeat.

2. Worked example: why the number $2/\eta$ appears

Take a one-direction quadratic $L(z) = \frac{\lambda}{2}z^2$ with $\lambda > 0$. Its gradient is λz, so one GD step gives

$$z_{t+1} = z_t - \eta\lambda z_t = (1 - \eta\lambda)z_t.$$

The distance shrinks exactly when $|1 - \eta\lambda| < 1$, equivalently $0 < \eta\lambda < 2$. For $\lambda = 5$ and $\eta = 0.3$, the multiplier is -0.5: each step changes sign but contracts. For $\lambda = 8$, it becomes -1.4: each step changes sign and expands.
What this does *not* prove. In a real network, the direction and curvature are not fixed. A step that expands one residual component may also change the weights enough to *lower* subsequent curvature. The paper's four-phase model seeks to explain this feedback.

3. The paper's mechanism: residual modes and changing weights

In the squared-loss linear-network setting, the authors write a vector of prediction residuals $D(t)$ and an effective Gram matrix $M^*(t)$, obtaining a discrete evolution of the form

$$D(t+1) = \big(I - \eta M^*(t)\big)D(t).$$

This resembles the scalar example, except that the matrix itself evolves with the weights. Project D onto the leading eigenvector v_1. Once its associated eigenvalue exceeds $2/\eta$, that component may grow geometrically, producing non-monotone loss. The weight updates then push sharpness down; when curvature falls enough, the component contracts again.

Why does the output layer matter?

The empirical observation is that changes in the *output-layer weight norm* often track changes in sharpness. The paper decomposes the network Gram matrix into contributions from output and inner-layer parameters. In its special linear model, this relation can be analyzed precisely; it is not an exact law for all deep nets.

4. Theoretical and empirical findings

Theorem 1 (Section 4.2, informal): Under a controlled eigengap/spectrum, sufficiently wide linear network, label alignment, and sample-size/width conditions, sharpness increases during an initial progressive-sharpening interval.
Theorem 2: With additional bounds on residual growth, sharpness, data spectrum and width, if sharpness rises sufficiently beyond $2/\eta$ it must later fall below a nearby threshold. A leading residual component can grow while above it. This is a *conditional* result for the special model.
Figures 1–2: Experiments show characteristic loss/sharpness oscillations and a lower-frequency relationship between output weight norm and sharpness. The authors openly note that fast sharpness oscillations and the roles of inner layers are not fully captured.

5. Proof idea: why a growing residual can reduce future sharpness

The top residual component obeys an approximate scalar recursion with multiplier $1 - \eta\Lambda_t$. In a frozen quadratic, $|1 - \eta\Lambda_t| > 1$ would make its magnitude grow forever. The key is that the network is *not frozen*: the same growing component changes the parameter gradients and can reverse the upward movement of sharpness. Theorem 2 formalizes parts of this feedback for a restricted linear model.
How to read Figure 1: Compare loss and sharpness timelines jointly. First identify when Λ crosses $2/\eta$; then find the delayed sharpness decrease and the nonmonotone loss interval. Oscillating sharpness alone does not show that optimization is improving. The authors highlight unexplained fast oscillations (Section 5).

5. Assumptions and limits

Assumptions: Full-batch, fixed-step GD; special two-layer linear architecture with symmetric initialization in the proof; restrictions on the spectrum of $X^\top X$, label projections, width/sample size and bounded residual behavior. The analysis uses squared error.
Limits: The four-phase interpretation combines experiments and partial heuristics; it is not a universal proof of EoS in nonlinear networks. The theoretical phase results do not establish generalization, optimal step-size choice, or convergence for arbitrary SGD. In particular, the norm/sharpness relationship can differ at high frequency.

6. Connections to the survey

Kalra (2025) later studies sharpness regimes via low-dimensional fixed points; **Chen–Bruna (2023)** characterizes stable two-step oscillations. This paper instead emphasizes *how sharpness evolves along a trajectory* and the four-phase feedback mechanism. **Kalra–Barkeshli (2024)** uses progressive sharpening and early sharpness reduction to understand warmup.

7. Quick reference

Concept	Meaning / equation
Sharpness	$\Lambda = \lambda_{\max}(\nabla^2 L)$
Quadratic boundary	$\eta\Lambda = 2$
Predicted residual dynamics	$D_{t+1} = (I - \eta M_t^*)D_t$
Four phases	below / above / decreasing / below threshold
Original to read	Sections 3.1–3.4, 4.2, Figure 1

8. Papers Master note

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Explains progressive sharpening and EoS as a four-phase curvature/residual feedback cycle. Formal sharpness claims require a restricted two-layer linear model and strong spectral/width assumptions; the full-network account remains partly empirical.

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